

FINAL PROJECT REPORT

ON

SMARTPHONE USAGE, DIGITAL ADDICTION & WELL-BEING ANALYTICS

A Data Analysis and Power Bi Visualization Project

Submitted by

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1. INTRODUCTION

1.1 Project Overview

The **Smartphone Usage, Digital Addiction & Well-Being Analytics** project is a data analytics and visualization project developed using **Microsoft Power BI**. The purpose of this project is to analyze smartphone usage patterns and understand their relationship with digital well-being, addiction, stress, sleep, social media usage, gaming, and work/study activities.

The dataset contains information about **7,500 users** and includes attributes such as age, gender, daily screen time, weekend screen time, social media hours, gaming hours, sleep hours, work/study hours, notifications per day, app opens per day, stress level, addiction level, and academic/work impact.

Power BI was used to transform the raw dataset into an interactive analytical report containing five report pages. The report uses KPI cards, bar charts, donut charts, line charts, scatter analysis, decomposition analysis, Key Influencers, and interactive filters.

The project provides both an overall summary and detailed analysis of smartphone usage behavior. It helps users explore different factors associated with smartphone addiction and digital well-being.

1.2 Objectives

The main objectives of this project are:

- To analyze smartphone usage patterns among 7,500 users.
- To calculate important overall KPIs related to smartphone usage.
- To analyze daily and weekend screen time.
- To examine social media and gaming usage.
- To analyze sleep and work/study hours.
- To compare smartphone usage across different genders.
- To compare usage patterns across different age groups.
- To analyze addiction levels and addicted status.
- To examine the relationship between stress and smartphone usage.
- To analyze factors associated with addicted status.
- To develop an interactive Power BI dashboard and report.
- To provide meaningful insights through data visualization.

2. PROJECT INITIALIZATION AND PLANNING PHASE

2.1 Define Problem Statement

Smartphones are widely used for communication, entertainment, social media, gaming, education, and work. However, excessive smartphone usage may be associated with changes in sleep, stress, productivity, and digital well-being.

The project dataset contains **7,500 user records** with several smartphone-related attributes. Since the dataset contains multiple variables, manually analyzing the information can make it difficult to identify important patterns.

Therefore, the problem addressed by this project is:

To analyze smartphone usage behavior and identify patterns related to smartphone addiction and digital well-being using interactive data visualization and business intelligence techniques.

The project aims to convert raw user data into meaningful visual information that can be easily understood and explored.

2.2 Project Proposal – Proposed Solution

The proposed solution is an interactive **Power BI-based Smartphone Addiction and Digital Well-Being Analytics Report**.

The project follows the following process:

Raw Data → Data Preparation → Data Exploration → Data Modelling → Calculations → Visualization → Dashboard → Insights

The Power BI report provides different analytical views covering:

- Overall user statistics
- Daily screen-time behavior
- Weekend screen-time behavior
- Addiction levels
- Addicted vs. non-addicted users
- Gender-wise analysis
- Age-group analysis
- Stress-level analysis
- Sleep analysis
- Social media usage

- Gaming usage
- App usage
- Notifications
- Work/study hours
- High screen-time users
- Key influencing factors

2.3 Initial Project Planning

Phase	Project Phase
Data Collection	Obtain and import smartphone usage dataset
Data Understanding	Examine fields, records and data types
Data Preprocessing	Check and prepare data for analysis
Data Analysis	Identify important usage and well-being variables
Data Modelling	Prepare fields for Power BI analysis
Calculations	Create required measures and analytical calculations
Visualization	Develop charts, KPI cards and analytical visuals
Dashboard Development	Create interactive Power BI pages
Testing	Test filters, visuals and interactions
Documentation	Prepare final project report

3. Data Collection and Preprocessing Phase

3.1 Data Collection Plan and Raw Data Sources Identified

The project uses a smartphone usage dataset containing information for **7,500 users**.

The major variables represented in the Power BI report include:

- User ID & Gender
- Age & Age Group

- Daily Screen Time Hours
- Weekend Screen Time
- Social Media Hours
- Gaming & Sleep Hours
- Work/Study Hours
- Notifications per Day
- App Opens per Day
- Addiction & Stress Level
- Academic/Work Impact
- Addicted Status
- High Screen-Time User status

The dataset was imported into Microsoft Power BI for analysis and visualization.

3.2 Data Quality Report

The dataset was reviewed before developing the dashboard. The analysis considered numerical, categorical, and classification-related fields.

The data preparation process included checking:

- User records
- Numeric fields
- Categorical fields
- Age groups
- Gender categories
- Addiction levels
- Stress levels
- Screen-time variables
- Sleep and work/study variables
- Missing or inconsistent values where applicable

The final report successfully uses the dataset for interactive analysis across multiple dimensions.

3.3 Data Exploration and Preprocessing

The data was explored to understand the relationships between smartphone usage and digital well-being.

Important derived or analytical categories used in the report include:

- **Age Group:** Under 20, 20–29, and 30–39
- **Addiction Level:** None, Mild, Moderate, Severe
- **Stress Level:** Low, Medium, High
- **Gender:** Male, Female, Other
- **Addicted Status:** Addicted / Not Addicted
- **High Screen-Time User:** High Usage / Normal Usage

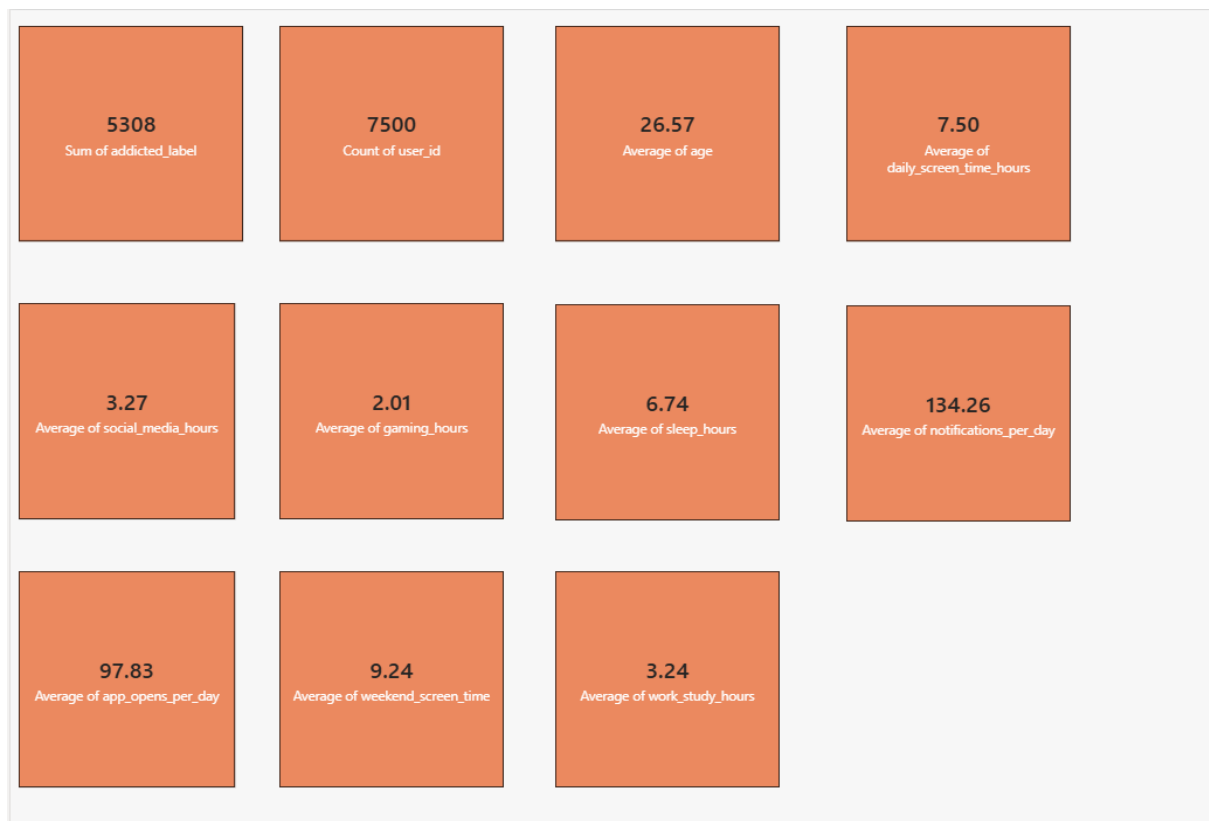
The data was then used to create different visualizations and analytical pages in Power BI.

4. DATA VISUALIZATION

4.1 Framing Business Questions

The following business/analytical questions were considered while developing the report:

1. How many users are included in the dataset?
2. What is the average age of the users?
3. What is the average daily screen time?
4. How much time do users spend on social media?
5. How much time do users spend gaming?
6. What is the average sleep duration?
7. How frequently do users open smartphone applications?
8. How many notifications do users receive per day?



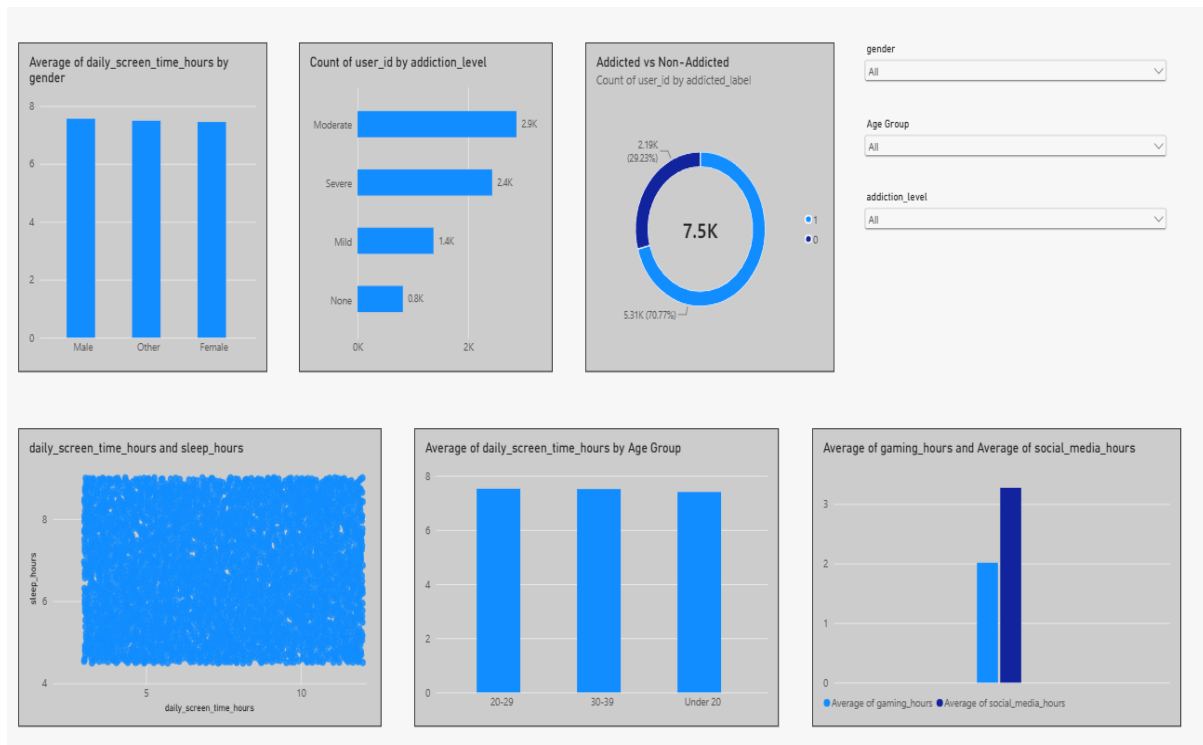
4.2 Developing Visualizations

Power BI was used to develop different types of visualizations based on the analytical requirements.

The report contains:

- KPI Cards
- Bar Charts
- Donut Charts
- Line Charts
- Scatter Plot
- Decomposition Tree
- Key Influencers
- Interactive Slicers

The visuals were arranged across five pages to provide a structured analytical experience.



5. DASHBOARD

5.1 Dashboard Design Screenshot with Insights

The project dashboard is designed around the theme:

Smartphone usage, Digital Addiction & Well-Being Analytics

The dashboard provides an overall view of smartphone usage using KPI cards and interactive charts.

Dashboard Insights

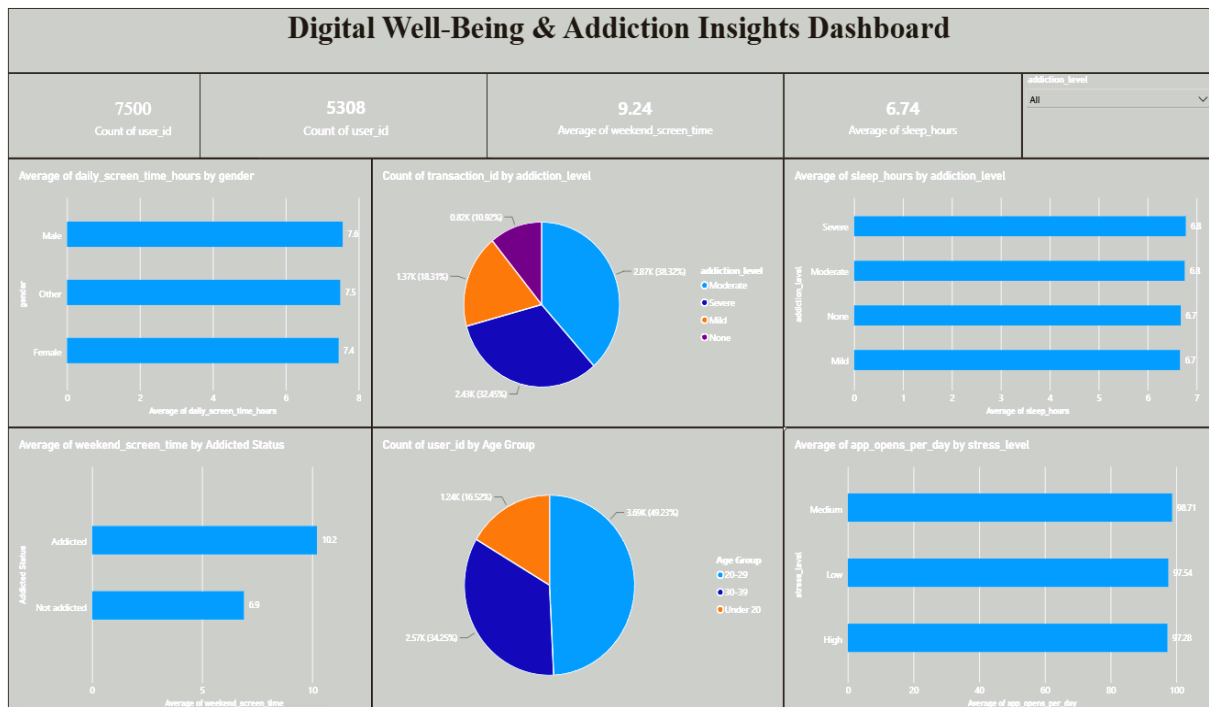
The dashboard indicates that the dataset contains **7,500 users**, with an average age of **26.57 years**.

The average daily screen time is **7.50 hours**, while average weekend screen time is higher at **9.24 hours**.

Users spend an average of **3.27 hours on social media** and **2.01 hours gaming**. The average sleep duration is **6.74 hours**, while the average work/study time is **3.24 hours**.

The average user also receives approximately **134.26 notifications per day** and opens applications approximately **97.83 times per day**.

These KPIs provide an overall picture of smartphone interaction within the dataset.



6. REPORT

6.1 Report Design Screenshot with Insights

The project contains **two detailed Power BI report pages** developed to provide deeper analysis of smartphone usage and digital well-being. These report pages complement the main dashboard by presenting detailed comparisons, trends, and relationships between different user characteristics.

6.1.1 Report 1 – Digital Addiction & Screen Time Behavior Analysis

The first report page focuses on smartphone addiction and screen-time behavior.

The report contains KPI cards for:

- Average app opens per day – **97.83**
- Average social media hours – **3.27**
- Average notifications per day – **134.26**

It also includes filters for **Addiction Level** and **Age Group**.

Major Visualizations

1. Addiction Level Distribution

The user distribution across addiction levels is approximately:

- Moderate – **2.9K**
- Severe – **2.4K**

- Mild – **1.4K**
- None – **0.8K**

This provides an overview of how users are distributed across different addiction categories.

2. Academic/Work Impact

Average work/study hours are approximately:

- No academic/work impact – **3.3 hours**
- With academic/work impact – **3.2 hours**

3. Weekend Screen Time by Addiction Level

The report shows higher average weekend screen time for users in higher addiction categories:

- Severe – **10.4 hours**
- Moderate – **10.1 hours**
- Mild – **7.3 hours**
- None – **6.2 hours**

4. Weekend Screen Time by Age Group

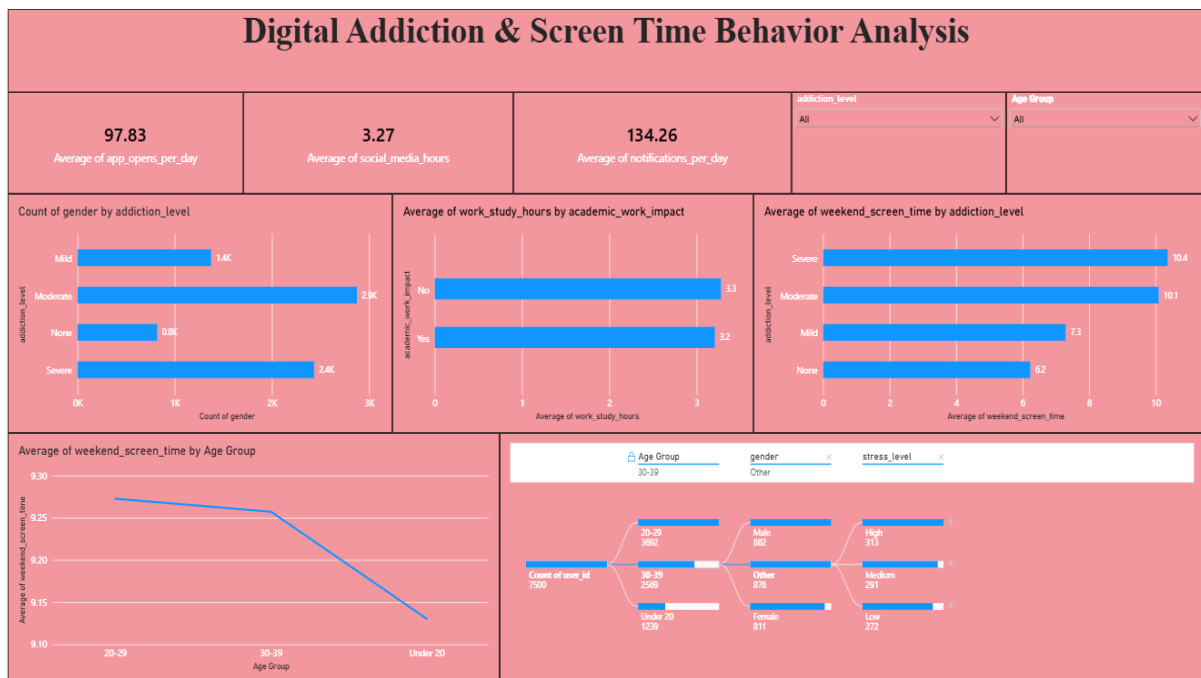
The line chart compares average weekend screen time among different age groups and allows users to observe differences between the **20–29, 30–39, and Under 20** groups.

5. Decomposition Tree

The decomposition tree provides an interactive breakdown of the **7,500 users** by:

Age Group → Gender → Stress Level

For example, the 20–29 age group contains approximately **3,692 users**.



6.1.2 Report 2 – Smartphone Addiction & Digital Well-Being Insights

The second report page focuses on digital well-being and factors associated with smartphone addiction.

The KPI cards display:

- Average Gaming Hours – **2.01**
- Average Age – **26.57 years**
- Average Work/Study Hours – **3.24 hours**

Filters are provided for **Gender** and **Stress Level**.

Major Visualizations

1. Stress Level Distribution

The 7,500 users are distributed across three stress categories:

- Low – approximately **34.13%**
- Medium – approximately **33.37%**
- High – approximately **32.49%**

This shows that the dataset contains users across all three stress categories.

2. Sleep Hours by Age Group

The line chart compares average sleep hours across age groups. The displayed averages are around **6.7 hours**, with differences between the age categories.

3. High Screen Time User Analysis

The donut chart compares high-usage and normal-usage categories. The displayed aggregate is approximately **56M**, divided into:

- High Usage – **66.12%**
- Normal Usage – **33.87%**

This metric represents the aggregation used in the Power BI visual rather than a direct count of unique users.

4. Social Media Usage by Addiction Level

Average social media usage increases across the displayed addiction categories:

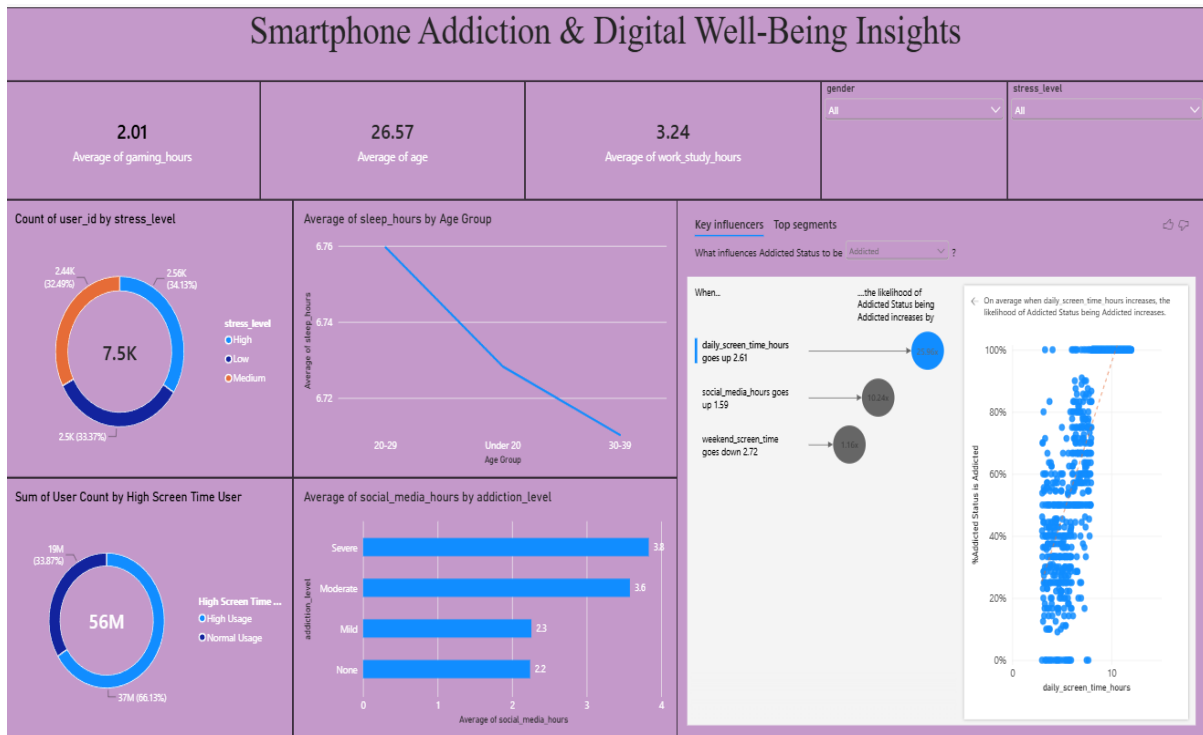
- Severe – **3.8 hours**
- Moderate – **3.6 hours**
- Mild – **2.3 hours**
- None – **2.2 hours**

5. Key Influencers

The Key Influencers visual examines factors associated with **Addicted Status**. The analysis highlights variables including:

- Daily screen time
- Social media hours
- Weekend screen time

These relationships should be interpreted as **associations within the dataset rather than proof that one factor causes addiction.**



7. PERFORMANCE TESTING

7.1 Utilization of Data Filters

Interactive slicers were implemented to improve report usability.

The report pages include filters such as:

- Addiction Level
- Age Group
- Gender
- Stress Level

When a filter is selected, the relevant charts and KPI values update accordingly. This allows users to analyze specific segments without changing the underlying dataset.

7.2 Calculation Columns / Measures

The project uses Power BI aggregations and calculated fields to derive meaningful analytical values.

Examples include:

- Average Age
- Average Daily Screen Time
- Average Social Media Hours

- Average Gaming Hours
- Average Sleep Hours
- Average Weekend Screen Time
- Average Work/Study Hours
- Average App Opens
- Average Notifications
- User counts by addiction level
- User counts by stress level

These calculations support the KPI cards and analytical visualizations.

7.3 Number of Visualizations

Across the **two detailed report pages**, the screenshots show approximately **15 analytical visuals**, excluding slicers and page-level controls.

These include KPI cards, bar charts, line charts, donut charts, a decomposition tree, and a Key Influencers visual.

8. CONCLUSION / OBSERVATION

The **Smartphone Addiction & Digital Well-Being Insights** project successfully converts smartphone usage data of 7,500 users into an interactive Power BI analytical solution.

The analysis provides visibility into screen time, addiction levels, social media usage, gaming, sleep, stress, notifications, app usage, and work/study patterns.

Some important observations from the report include:

- Average daily screen time is **7.50 hours**.
- Average weekend screen time is **9.24 hours**.
- Average social media usage is **3.27 hours per day**.
- Average gaming usage is **2.01 hours per day**.
- Average sleep is **6.74 hours**.
- The largest displayed addiction category is **Moderate**, followed by **Severe**.
- Higher addiction categories show higher average weekend screen time in the dataset.

- Social media usage is higher in the Severe and Moderate addiction categories than in the Mild and None categories.
- The dataset contains a relatively balanced distribution across Low, Medium, and High stress categories.

Overall, Power BI provides an effective way to explore these patterns interactively and present the results in an understandable format.

9. FUTURE SCOPE

The project can be further enhanced in the following ways:

- Add historical data to analyze changes in smartphone usage over time.
- Include additional behavioral and lifestyle variables.
- Develop predictive models for identifying users at risk of excessive smartphone use.
- Add machine-learning-based addiction prediction.
- Create personalized digital well-being recommendations.
- Add mobile-friendly dashboard layouts.
- Connect the dashboard to live or regularly updated data sources.
- Introduce automated alerts for unusually high screen-time patterns.

10. APPENDIX

10.1 Source Code

The project was primarily developed using **Microsoft Power BI**. Data transformation and analysis were performed using Power Query and Power BI calculations.

No separate application source code was required for the dashboard development.

Tools used:

- Microsoft Power BI
- Power Query
- DAX/Power BI calculations
- Dataset containing 7,500 user records

10.2 GitHub & Project Demo Link

GitHub Repository:

<https://github.com/skarifun00-droid>

Project Title: Smartphone Addiction & Digital Well-Being Analytics

Developed using Microsoft Power BI

Dataset: 7,500 Users

Dashboard: 1

Detailed Report Pages: 2

END OF PROJECT REPORT